8.4: Modular Arithmetic with Applications to Cryptography

Definition.
The **Caesar cipher** encrypts messages by shifting letters by a fixed amount (typically 3 places). The encrypted messages are decrypted by shifting in the opposite direction.

Example. Use a Ceasar cipher with a shift of 3 to

encrypt the message MATH RULES

	13	01	20	08	18	21	12	05	19
+3	16	04	23	11	21	24	15	08	22
	P	D	W	K	U	X	O	H	V

A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

decrypt the message ODQGHU

	15	04	17	07	08	21
-3	12	01	14	03	05	18
	L	A	N	D	E	R

Definition.
Given integers a and n with $n > 1$, the **residue of a modulo n** is $a \mod n$, the non-negative remainder obtained when a is divided by n.

- The numbers $0, 1, \ldots, n-1$ are called a **complete set of residues modulo n.**
- **Reducing a number modulo** n means to set it equal to its residue modulo n.
- If a modulus $n > 1$ is fixed throughout a discussion and an integer a is given, the words "modulo n" are often dropped and we simply speak of **the residue of a.**

Theorem 8.4.2: Congruence Modulo n is an Equivalence Relation
If n is any integer with $n > 1$, congruence modulo n is an equivalence relation on the set of all integers. The distinct equivalence classes of the relation are the sets $[0], [1], [2], \ldots, [n-1]$, where for each $a = 0, 1, 2, \ldots, n-1$,

$$\begin{aligned}[a] &= \{m \in \mathbb{Z} \mid m \equiv a \mod n\} \\ &= \{m \in \mathbb{Z} \mid m = a + kn \text{ for some integer } k\}\end{aligned}$$

Theorem 8.4.3: Modular Arithmetic
Let a, b, c, d, and n be integers with $n > 1$, and suppose

$$a \equiv c \mod n \qquad \text{and} \qquad b \equiv d \mod n.$$

Then

1. $(a + b) \equiv (c + d) \mod n$
2. $(a - b) \equiv (c - d) \mod n$
3. $ab \equiv cd \mod n$
4. $a^m \equiv c^m \mod n$, for every positive integer m.

Corollary 8.4.4
Let a, b and n be integers with $n > 1$. Then

$$ab \equiv [(a \bmod n)(b \bmod n)] (\bmod n)$$

or, equivalently,

$$ab \mod n = [(a \bmod n)(b \bmod n)] (\bmod n)$$

In particular, if m is a positive integer, then

$$a^m \equiv [(a \bmod n)^m] (\bmod n).$$

Example. Compute the following

$55 + 26 \mod 4$

$= 55 \pmod 4 + 26 \pmod 4$

$= 3 \pmod 4 + 2 \pmod 4$

$= 5 \pmod 4$

$= 1 \pmod 4$

$55 - 26 \mod 4$

$= 55 \pmod 4 + 26 \pmod 4$

$= 3 \pmod 4 - 2 \pmod 4$

$= 1 \pmod 4$

$55 \cdot 26 \mod 4$

$= 55 \pmod 4 \cdot 26 \pmod 4$

$= 3 \pmod 4 \cdot 2 \pmod 4$

$= 6 \pmod 4$

$= 2 \pmod 4$

$55^2 \mod 4$

$= 55^2 \pmod 4$

$= 3^2 \pmod 4$

$= 9 \pmod 4$

$= 1 \pmod 4$

Recall the following property of exponents:

- $x^{2a} = (x^a)^2$
- $x^{a+b} = x^a x^b$

for all real numbers x, a, and b with $x \geq 0$. By Corollary 8.4.4,

$$x^7 \mod n = \left[(x^4 \bmod n)(x^2 \bmod n)(x^1 \bmod n)\right] \mod n.$$

Example. Compute the following:

$144^4 \mod 713$

$= (144^2)^2 \pmod{713}$

$= 20736^2 \pmod{713}$

$= 59^2 \pmod{713}$

$= \boxed{629 \pmod{713}}$

$12^{43} \mod 713$

$43 = 32 + 8 + 2 + 1$

$12 \pmod{713} = 12 \pmod{713}$

$12^2 \pmod{713} = 144 \pmod{713}$

$12^4 \pmod{713} = 59 \pmod{713}$

$12^8 \pmod{713} = 629 \pmod{713}$

$12^{16} \pmod{713} = 639 \pmod{713}$

$12^{32} \pmod{713} = 485 \pmod{713}$

$\Rightarrow 12^{43} \bmod 713 = 12^{32} \cdot 12^8 \cdot 12^2 \cdot 12^1 \pmod{713}$

$= 485 \cdot 629 \cdot 144 \cdot 12 \pmod{713}$

$= 527152320 \pmod{713}$

$= \boxed{48 \pmod{713}}$

Definition.
An integer d is said to be a **linear combination of integers** a and b if, and only if, there exists integers s and t such that $as + bt = d$.

Theorem 8.4.5: Writing a Greatest Common Divisor as a Linear Combination
For all integers s and b, not both zero, if $d = \gcd(a, b)$, then there exist integers s and t such that $as + bt = d$.

Example. Use the extended Euclidean algorithm to express $\gcd(660, 43)$ as a linear combination of 660 and 43.

div mod

$$660 = 43 \cdot 15 + 15$$
$$43 = 15 \cdot 2 + 13$$
$$15 = 13 \cdot 1 + 2$$
$$13 = 2 \cdot 6 + 1$$
$$2 = 1 \cdot 2 + 0$$

$$\boxed{\gcd(660, 43) = 1}$$

$$\begin{aligned}
1 &= 13 - 2 \cdot 6 \\
&= 13 - (15 - 13 \cdot 1) \cdot 6 \\
&= 7 \cdot 13 - 6 \cdot 15 \\
&= 7 \cdot (43 - 15 \cdot 2) - 6 \cdot 15 \\
&= 7 \cdot 43 - 20 \cdot 15 \\
&= 7 \cdot 43 - 20 \cdot (660 - 43 \cdot 15) \\
&= \boxed{307 \cdot 43 - 20 \cdot 660}
\end{aligned}$$

Definition.
Given any integer a and any positive integer n, if there exists an integer s such that $as \equiv 1(\mod n)$, then s is called **an inverse for a modulo n**.
Integers a and b are **relatively prime** if, and only if, $\gcd(a, b) = 1$.

Corollary 8.4.6
If a and b are relatively prime integers, then there exist integers s and t such that $as + bt = 1$.

Corollary 8.4.7
For all integers a and n, if $\gcd(a, n) = 1$, then there exists an integer s such that $as \equiv 1(\mod n)$, and so s is an inverse for a modulo n.

Example.

Find an inverse for 43 modulo 660. (Solve $43s \equiv 1(\mod 660)$.)

Recall: $1 = 307 \cdot 43 - 20 \cdot 660$

$\Rightarrow 43(307) = 1 + 20 \cdot 660$

$\Rightarrow 43(307) \equiv 1 \pmod{660}$

Find a positive inverse for 3 modulo 40. (Solve $3s \equiv 1(\mod 40)$.)

div mod

$40 = 3 \cdot 13 + 1 \Rightarrow 3(-13) = 1 + (-1)40$

$\Rightarrow 3(-13) \equiv 1 \pmod{40}$

$\Rightarrow 3(27) \equiv 1 \pmod{40}$

Least positive inverse

Definition. (RSA Cryptography)
RSA Encryption, named after Rivest, Shamir, and Adleman, uses modular arithmetic to encrypt and decrypt a plaintext message, M, into a ciphertext, C.
Given prime integers p and q, and a positive integer e that is relatively prime to $(p-1)(q-1)$,

- messages are encrypted using the formula:

$$C = M^e \mod pq.$$

  The number pair (pq, e) is a **public key**, which is shared.

- messages are decrypted using the formula:

$$M = C^d \mod pq$$

  The number pair (pq, d) is a **private key**. The integer d is a positive inverse to e modulo $(p-1)(q-1)$.

The security of RSA encryption relies on the difficulty of factoring very large numbers.

Example. Let $p = 5$ and $q = 11$. Since $(p-1)(q-1) = 40$, we may choose $e = 3$, which is relatively prime to 40.

A	1	N	14
B	2	O	15
C	3	P	16
D	4	Q	17
E	5	R	18
F	6	S	19
G	7	T	20
H	8	U	21
I	9	V	22
J	10	W	23
K	11	X	24
L	12	Y	25
M	13	Z	26

Encrypt the plaintext "HI"

08 09

$$C = 08^3 \bmod 55 = 64 \cdot 8 \bmod 55 = 9 \cdot 8 \bmod 55 = 72 \bmod 55 = 17$$

$$C = 09^3 \bmod 55 = 81 \cdot 9 \bmod 55 = 26 \cdot 9 \bmod 55 = 234 \bmod 55 = 14$$

Decrypt the ciphertext "17 14"

<u>Recall</u> $3(27) \equiv 1 \pmod{40} \Rightarrow d = 27$

$$M = 17^{27} \pmod{55} = 17^{16} \cdot 17^{8} \cdot 17^{2} \cdot 17 \pmod{55} = 8$$

$$M = 14^{27} \pmod{55} = 14^{16} \cdot 14^{8} \cdot 14^{2} \cdot 14 \pmod{55} = 9$$

Algorithm 1: Algorithm 8.4.1: Extended Euclidean Algorithm

Input : A, B [integers with $A > B > 0$]

$a := A,\ b := B,\ s := 1,\ t := 0,\ u := 0,\ v := 1$

while $b \neq 0$ **do**

  $r := a \bmod b, \qquad q := a \operatorname{div} b$

  $a := b, \qquad b := r$

  $u_{\text{new}} := s - uq, \qquad v_{\text{new}} := t - vq$

  $s := u, \qquad t := v$

  $u := u_{\text{new}}, \qquad v := v_{\text{new}}$

end

$\text{gcd} := a$

Output: gcd [a positive integer], s, t [integers]

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a=660, b=43, s=1, t=0, u=0, v=1
----------------------------------------
r=15                    q=15
a=43                    b=15
unew=1-0·15=1           vnew=0-1·15=-15
s=0                     t=1
u=1                     v=-15
----------------------------------------
r=13                    q=2
a=15                    b=13
unew=0-1·2=-2           vnew=1-(-15)·2=31
s=1                     t=-15
u=-2                    v=31
----------------------------------------
r=2                     q=1
a=13                    b=2
unew=1-(-2)·1=3         vnew=-15-31·1=-46
s=-2                    t=31
u=3                     v=-46
----------------------------------------
r=1                     q=6
a=2                     b=1
unew=-2-3·6=-20         vnew=31-(-46)·6=307
s=3                     t=-46
u=-20                   v=307
----------------------------------------
r=0                     q=2
a=1                     b=0
unew=3-(-20)·2=43       vnew=-46-307·2=-660
s=-20                   t=307                          gcd(660,43)=a=1
u=43                    v=-660                         s=-20, t=307
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